

# AMAN KUMAR PANDEY

✉ paman7681@gmail.com 📞 +91 6207076733 🔗 linkedin.com/in/amanpandey2101 🐙 github.com/amanpandey2101

## TECHNICAL SKILLS

|                           |                                                                               |
|---------------------------|-------------------------------------------------------------------------------|
| Languages                 | Python, Java, C++, Go, TypeScript, SQL                                        |
| Cloud & DevOps            | AWS, GCP, Docker, Terraform, Linux, Networking, Git, CI/CD                    |
| Backend & Frontend        | FastAPI, Flask, Node.js, Next.js, React, REST APIs, Microservices             |
| Testing & Methodology     | Unit Testing, Code Reviews, Debugging, Agile/Scrum, Prometheus, Grafana       |
| AI, Databases & Messaging | GenAI, LangGraph, RAG, PostgreSQL, MongoDB, DynamoDB, Redis, RabbitMQ, Celery |

## EXPERIENCE

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| <b>LiveSitter</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Oct 2024 – Present          |
| <i>Software Development Engineer (Contract)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <i>Atlanta, GA (Remote)</i> |
| <ul style="list-style-type: none"><li>Architected a <b>scalable, multi-tenant enterprise platform</b> using Next.js 16 and Flask, implementing AI-driven document processing, classification, and compliance tracking with a custom template engine to <b>reduce manual review time by 80%</b>.</li><li>Engineered an autonomous <b>LangGraph</b>-powered conversational agent and <b>RAG (FAISS)</b> pipelines integrated into the admin support portal to automate candidate messaging and text-based screening.</li><li>Developed Python-based edge inference software on <b>Nvidia Jetson Orin</b> running local YOLO and RT-DETR models; routed critical alerts via local network or cloud push notifications and SMS to the customer.</li><li>Deployed backend services on <b>AWS ECS Fargate</b> behind ALBs with auto-scaling; implemented secure credential workflows via Secrets Manager, Lambda functions, and automated S3/database cleanup cron jobs.</li><li>Built self-hosted <b>CI/CD pipelines</b> to meet SOC2 compliance standards, and deployed a <b>Prometheus and Grafana</b> observability stack to monitor system load, inferencing pipelines, and worker health.</li><li>Led <b>code reviews, PR reviews, and unit testing</b> across the full stack in an Agile workflow; collaborated cross-functionally with product, design, and QA to define solutions and estimate scope of work.</li><li>Engineered a Go-based <b>on-demand snapshot API</b> that downsampled 30 FPS RTSP inputs to 1 FPS via FFmpeg keyframe decoding, <b>cutting CPU usage by 30%</b> and scaling to <b>40+ cameras per edge box</b>.</li><li>Engineered a <b>fault-tolerant C++ orchestration layer</b> using the Sophgo SDK, featuring self-healing process supervisors and auto-restart logic to ensure <b>99.9% uptime</b> at the hardware edge.</li></ul> |                             |

## PROJECTS

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| <b>Memora.dev: AI Decision Intelligence Platform</b>   <i>Next.js, FastAPI, AWS (ECS Fargate, Bedrock, DynamoDB)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>GitHub</b>                      |
| <ul style="list-style-type: none"><li>Engineered an event-driven platform utilizing <b>FastAPI, SQS, and DynamoDB</b> to parse and analyze GitHub commit and PR diffs, auditing source code changes to capture and preserve engineering trade-offs.</li><li>Architected a RAG-based AI pipeline utilizing <b>Amazon Bedrock</b> and <b>OpenSearch</b> to generate structured Architectural Decision Records (ADRs) and explain complex design decisions with evidence-backed annotations.</li><li>Deployed auto-scaling compute environments using <b>Docker containers</b> running on AWS ECS Fargate, configuring network access logs, target group definitions, and healthy state checking policies.</li></ul> |                                    |
| <b>Multi Agent Framework &amp; Software Development Platform</b>   <i>Python, FastAPI, Next.js, Prisma, PostgreSQL</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>GitHub</b>                      |
| <ul style="list-style-type: none"><li>Developed a multi-agent system designed to automatically generate, test, and <b>evaluate code correctness</b>; features specialized agents that perform <b>static code analysis</b>, linting, and bug detection.</li><li>Implemented DAG-based task scheduling, shared memory, secure tool execution, and multi-model routing to coordinate specialized AI agents efficiently.</li><li>Engineered a <b>preference-based code ranking algorithm</b> where specialized review agents evaluate code efficiency, clean architecture, and correctness to select optimal solutions.</li></ul>                                                                                     |                                    |
| <b>HAKS: Digital Welfare Infrastructure for Martyrs' Families</b>   <i>MERN, React Native</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <a href="#">[View Recognition]</a> |
| <ul style="list-style-type: none"><li>Architected a <b>cross-platform digital ecosystem</b> (Web/Mobile) to streamline record management and benefit disbursement for the Indian Army, replacing manual legacy workflows.</li><li>Engineered an <b>automated document synthesis engine</b> to generate pre-filled welfare forms, reducing grievance filing latency and improving data accuracy for stakeholders.</li><li>Successfully deployed the portal in collaboration with the <b>Bihar Regiment</b>, receiving official recognition for enhancing digitized welfare access and grievance redressal efficiency.</li></ul>                                                                                    |                                    |

## ACHIEVEMENTS & LEADERSHIP

- First Runner-Up Winner in **AI for Bharat by Amazon Web Services** 2026 Hackathon. [\[View Recognition\]](#)
- Awarded by the **Indian Army** for the HAKS project impact.
- Led **200+** students as Head of ASET AIKYAM, organizing technical workshops and hackathons.

## EDUCATION

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|-------------------------------------------------------------------|-----------------------|
| <b>Amity University, Patna</b>                                    | June 2022 – June 2026 |
| <i>Bachelor of Technology in Computer Science and Engineering</i> | <i>CGPA: 8.89</i>     |